

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> To find the current drawn by the head light bulb , we can use the formula $P = V \times I$, where P is the power dissipated , V is the voltage , and I is the current . Rearranging the formula to solve for I , we get $I = P / V$. Substituting the given values , $I = 40 \text{ watts} / 12 \text{ volts}$. This calculation yields the current drawn by the bulb . Then , to find the resistance , we can use the formula $R = V / I$, substituting the calculated current and given voltage values . This will give us the resistance of the bulb . </think><answer> 3 . 33 amps </answer>

